

Open-Source LLMs — Setup Checklist (Day 2 · v3)

Send this to students 48-72 hours before class.

Day 2 is keyboard-first: you'll type your first command in **minute one** of class, and your laptop will run a real LLM by **minute 20**. That only works if you do this 15-minute setup at home first. Skip it, and you'll spend the first labs watching other people have fun.

TL;DR — what you need by class start

1. **Ollama installed** — and verified with `ollama --version`. **Most important.**
2. **One hosted open-provider account** — Groq, Together AI, or Hugging Face (pick one, free).
3. **Python 3.10+** with the `requests` package — for the big lab.
4. **A laptop with 8+ GB RAM and 5 GB free disk.**

Bonus (saves you class time): pull the class model in advance — `ollama pull qwen3:0.6b`.

Step 1 — Install Ollama and VERIFY it

This is the most important step. Class literally opens with this check.

Install

- **Mac:** download from ollama.com/download — drag-and-drop install.
- **Windows:** download the installer from ollama.com/download — runs as a service. Use **PowerShell** for the commands below (not CMD, not WSL — unless you installed inside WSL).
- **Linux:** one-liner — `curl -fsSL https://ollama.com/install.sh | sh`

Verify (the exact check we run in minute one of class)

Open a terminal (Mac: Terminal · Windows: PowerShell) and type:

```
ollama --version
```

A version number = you're set. An error = restart the terminal first (the installer edits PATH); if it still fails, see Troubleshooting below and message the class channel BEFORE class.

Optional but smart — pull the class model now

```
ollama pull qwen3:0.6b
```

~520 MB. We'll do this together in class over classroom wifi — but if you pull it at home, you skip the wait entirely.

16 GB laptop? Also grab a mid-size model for the lab's stretch goal:

```
ollama pull mistral:7b
```

(~4.4 GB — definitely nicer to do on home wifi.)

Step 2 — Sign up for ONE hosted open provider

For the cross-model bake-off (first lab), you need a free account on **one** of:

- **Groq** — console.groq.com — **recommended**: fastest signup (Google login), no credit card, very fast inference.
- **Together AI** — together.ai — most model variety, free credit on signup.
- **Hugging Face** — huggingface.co — largest catalog, free Inference API.

After signup, **log in once and make sure you can reach the model playground/console**. That's all — no API key needed for the bake-off (we use the web console).

Step 3 — Python check (for the big lab)

```
python --version      # need 3.10+  (or python3 --version)
pip install requests  # (or pip3)
```

That's the only package the lab uses. If you have Jupyter or VS Code, great — but a plain terminal works fine.

One-command check (recommended): if you received the handout pack, run:

```
python homework/examples/verify_setup.py
```

It checks Ollama + the class model in one shot and tells you exactly what to fix.

Step 4 — Hardware self-check (30 seconds)

Find your row — this is the model you'll use in the lab:

Your laptop	Your lab model
8 GB RAM, no GPU	qwen3:0.6b (the class default — works everywhere)
16 GB RAM	mistral:7b or llama3.1:8b (pull at home!)
16 GB + gaming GPU	up to 13B models
Apple Silicon 16+ GB	up to 13B–34B — unified memory punches above its weight

Small laptop? Zero problem. qwen3:0.6b runs the entire class. You will not be left out.

Ollama command reference (keep this handy in class)

The eight commands you'll actually use — everything else is rare.

Everyday

```
ollama pull <model>      # download a model    e.g. ollama pull qwen3:0.6b
ollama run <model>       # chat in the terminal (auto-pulls if missing)
ollama list              # see what's installed + sizes on disk
ollama ps               # see what's RUNNING right now (loaded in RAM)
```

Housekeeping

```
ollama rm <model>          # DELETE a model, free the disk space
                             # e.g. ollama rm mistral:7b
ollama stop <model>         # unload a running model from RAM (keeps the file)
ollama show <model>         # model card: params, quantization, context size, license
```

The server (for the Python lab)

```
ollama serve                # start the API server on localhost:11434
                             # (usually already running as a background service –
                             # if you get "address already in use", that's GOOD, skip it)
```

Inside ollama run chat

Type	Does
/bye	exit the chat
/clear	wipe the conversation (fresh context window — Day 1 concept, live!)
/show info	model details without leaving chat
/set system <text>	set a SYSTEM prompt right in the terminal

Two useful checks

```
ollama list                 # before class: is qwen3:0.6b there?
du -sh ~/.ollama            # Mac/Linux: total disk all models use
                             # Windows: check size of C:\Users\<you>\.ollama
```

Disk hygiene tip: models add up fast — a few 7B pulls = 20+ GB. `ollama list` shows sizes; `ollama rm` reclaims space instantly. Deleting a model is safe — you can always pull it again.

Troubleshooting

Symptom	Fix
ollama: command not found after install	Close and reopen the terminal (PATH refresh). Windows: log out/in if needed.
ollama --version works but pull hangs	Corporate VPN/proxy often blocks the registry. Try home network, or grab the GGUF from a classmate via USB in class.
Windows: which terminal?	PowerShell. Not CMD. WSL only if you installed Ollama inside WSL.
“model requires more memory”	Your model is too big for your RAM — use <code>qwen3:0.6b</code> .
Corporate laptop, installs blocked	Message the instructor BEFORE class — the hosted-provider fallback covers every lab (same API shape).

What you do NOT need

- No OpenAI/Anthropic API key.
- No credit card anywhere.
- No GPU. (Nice if you have one; required never.)
- No Docker, no WSL, no cloud account beyond the one free provider login.

See you in class. First command at minute one — be ready.